

Topic: Google PageRank Algorithm and Website Evaluation

Required Content

1. Model the web system as a directed graph $G(V, E)$.
2. Construct the link matrix P and the Google matrix

$$G = \alpha P + (1 - \alpha) \frac{1}{n} \mathbf{ee}^T, \quad 0 < \alpha < 1.$$

3. Present the meaning of the PageRank equation:

$$\pi = G\pi.$$

4. Apply the PageRank algorithm to a dataset containing at least 100 web pages to compute rankings.
5. Analyze the influence of the damping factor α on ranking results.
6. Study several PageRank variants:
 - Personalized PageRank;
 - Topic-sensitive PageRank;
 - Weighted PageRank.

Description

- **Object:** A web graph with at least 100 pages (real or self-constructed).
- **Purpose:** Evaluate the importance of web pages based on link structure.
- The main steps of the system include:
 - Constructing the web graph;
 - Building the probability matrix;
 - Computing the ranking vector;
 - Comparing page rankings.
- Key concepts from the course must be clearly explained, including probability matrices, eigenvalues and eigenvectors, and Markov matrices.

Expected Results

- Develop a program to compute PageRank for a web graph with more than 100 pages.
- Produce a ranking table of web pages.
- Provide comments on the impact of link structure and the damping factor α .

Notes

- Do not use built-in PageRank libraries.